A FLAT SURJECTIVE MORPHISM FROM A REDUCED ALGEBRAIC SPACE HAS REDUCED TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one Properties of Spaces section and one Descent lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Definition of reduced algebraic spaces. Stacks Project, Properties of Spaces, Section spaces-properties-section-types-properties. Reduces reducedness to the \'etale charts.
- Scheme case. Stacks Project, Descent, Lemma descent-lemma-descend-reduced. Concludes for $U \to V$.

Source reference. Original reference 03E5: sources/stacks-chow/tags.txt, source lines 4444–4444. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06QM: sources/stacks-chow/tags.txt, source lines 8611–8611. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If f is flat and surjective and X is reduced, then Y is reduced.*

Proof. Choose a scheme V and a surjective étale morphism $V \to Y$. Choose a scheme U and a surjective étale morphism $U \to X \times_Y V$. As f is surjective and flat, the morphism of schemes $U \to V$ is surjective and flat. In this way we reduce the problem to the case of schemes (as reducedness of X and Y is defined in terms of reducedness of U and V, see Properties of Spaces, Section 03E5). The case of schemes is Descent, Lemma 06QM. □

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